2021 Ph H2 Q13

Section: Electricity

Topic: Current, PD, Power, Resistance

This question involves electrical power and resistance in a circuit.

(a)
Calculate the resistance of the lamp

Given:

- $V = 3.0\,\mathrm{V}$
- $I = 0.20\,\mathrm{A}$

Use Ohm's Law:

$$R = \frac{V}{I} = \frac{3.0}{0.20} = 15\,\Omega$$

(b)
Calculate the power output of the lamp

Use:

$$P = IV = 0.20 \times 3.0 = 0.60\,\mathrm{W}$$

(c)
Determine whether the resistor is operating at its rated power

Given:

- Resistor rating: $2.2\,\mathrm{k}\Omega$, $0.25\,\mathrm{W}$
- Voltage across resistor: $V = 24 - 3 = 21\,\mathrm{V}$

Power dissipated:

$$P = \frac{V^2}{R} = \frac{21^2}{2200} = \frac{441}{2200} \approx 0.20\,\mathrm{W}$$

✅ Since $0.20\,\mathrm{W} < 0.25\,\mathrm{W}$, the resistor is operating **within** its rated power.

🧠 **Revision Tips**

- Use $R = \frac{V}{I}$, $P = IV$, and $P = \frac{V^2}{R}$ depending on known values
- Always **subtract voltage drops** across components when analysing part of a circuit
- Rated power = maximum safe power dissipation
- If you exceed rated power, components may **overheat or fail**